

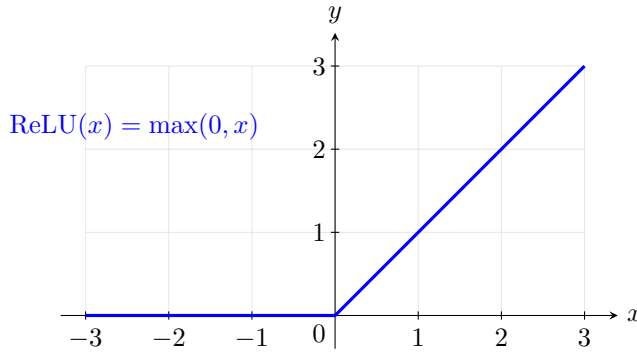
Master's Thesis Summary

Deep learning and o-minimality

Backpropagation and nonsmoothness

A routine training of a neural network is essentially an attempt to minimize $L(\theta) = \frac{1}{N} \sum_{i=1}^N l_i(\theta)$, where θ contains the network's parameters and l_i is the loss for example i . To minimize this average loss L , we need to determine which parameters of the network to change and by how much. This is done through backpropagation: we evaluate the derivatives of the operations involved in the network's architecture and combine them using the chain rule.

A key part of this architecture are the activation functions. One common example is ReLU, which is, of course, smooth everywhere but at 0, where it is non-differentiable. To backpropagate, we must be able to compute the derivative at every point, so some kind of convention is usually adopted; maybe the most obvious solution is to set the derivative at 0 to $\frac{1}{2}$.



These conventions can quickly become problematic if not used carefully: take $g(x) := x + 4 \text{ReLU}(-\text{ReLU}(x))$. If we backpropagate using the above convention for ReLU, we get $Dg(0) = 1 + 4 \left(\frac{1}{2}\right) \left(-\frac{1}{2}\right) = 0$. However, if we simply notice that $\text{ReLU}(x) \geq 0$ for all x , we see that $g(x) = x$ for all x . Therefore, the value at 0 proposed by the convention actually disagrees with the derivative of the represented function, although that function is smooth.

This sort of inconsistency is actually fairly problematic in the case of backpropagation, since functions are regularly composed with other functions and differentiated via the chain rule, so a function like g could easily appear somewhere in the process. Setting the ReLU derivative as above would then cause weird behavior when updating the weights and doing gradient descent. For example, a zero output from backpropagation can produce a zero gradient descent update at a point where the actual derivative is nonzero.

And there is no easy solution to this problem: for example, one might argue that this shouldn't be particularly problematic, as the common activation functions are differentiable almost everywhere anyway. However, intermediate operations can map whole regions of inputs into these sets of measure zero: ReLU itself maps all the nonpositive inputs to 0. Furthermore, training can approach nonsmooth boundaries, so behavior near them matters even if no iterate lies exactly on a nonsmooth boundary.

Several different frameworks of consistent conventions have been proposed, but almost all of them do not apply to all the commonly used activation functions. For example, a consistent way of assigning derivatives exists for convex functions and another framework deals with semialgebraic functions, which are essentially functions that can be built up out of polynomials in a certain way. These frameworks would cover ReLU, but not, for example, the logistic sigmoid activation function.

O-minimal geometry and the thesis project

Tools from o-minimal geometry, a branch of model theory, which is itself a subfield of mathematical logic, can be used to describe a huge class of functions, significantly larger than the previously proposed frameworks. The framework is in fact a generalized version of the semialgebraic framework, and has been estimated [1] to include around 89% of the commonly used activation functions. O-minimal geometry essentially deals with sets and functions definable by logical formulas over certain structures with nice properties. This ends up excluding many pathological functions and allows strong results to be developed on the functions that are included.

For functions of this framework that are also locally Lipschitz, *conservative fields* [2] provide a generalized derivative model that can include the outputs of nonsmooth backpropagation. These fields assign sets of vectors whose line integrals along parameter paths recover changes in the objective. Their composition and sum rules allow them to model computational graphs and minibatch updates. Under suitable assumptions on the learning rates and minibatch sampling, and assuming the parameter iterates remain bounded, this framework yields almost-sure convergence of the loss values and stationarity of every accumulation point with respect to the associated conservative field.

In the thesis, I will cover the necessary theoretical background, including the big

result(s) from o-minimal geometry which underpin the results discussed above. I will then apply these results and this framework to deep learning in the above-mentioned way and cover some of the applied results mentioned above.

References

- [1] Gilles Bareilles, Allen Gehret, Johannes Aspman, Jana Lepšová, and Jakub Mareček. Deep Learning as the Disciplined Construction of Tame Objects. arXiv:2509.18025v2, 2025.
- [2] Jérôme Bolte and Edouard Pauwels. Conservative set valued fields, automatic differentiation, stochastic gradient methods and deep learning. arXiv:1909.10300v4, 2020.